

The Bloom's-AI Integration Model

Lesson planning templates for science courses in the era of generative AI

The Bloom's-AI Integration Model

Lesson planning templates for science courses in the era of generative AI

Version 1.0 · Saroj Lamichhane · Licensed CC BY 4.0

The problem this addresses

A student asks a language model to balance an equation, name a compound, or explain a mechanism. The output is pasted in. The instructor grades work that reveals nothing about what the student understands.

Institutions have largely split into two responses, and both are failing.

Prohibition and detection. Unenforceable, unreliable, and it makes the instructor an adversary of the class. Detection tools produce false positives that fall hardest on students writing in a second language.

Permission without structure. Produces submissions that measure nothing, and teaches students that the tool does the thinking.

There is a third option, and it does not require the instructor to police anyone.

The principle

Change what is being assessed.

If the graded artifact is the answer, a model can produce it. If the graded artifact is the student's *engagement with* an answer — finding its error, adapting it to a new constraint, judging it against criteria — the model cannot produce that, because it requires the student to hold the chemistry in their head.

The model becomes the object of study rather than the source of the submission.

This is not a trick to catch students. It is a redesign that makes AI use productive, and it works whether or not a given student uses AI at all.

Mapping to Bloom's Taxonomy

Each level of the taxonomy suggests a different structure for an AI-integrated task.

Level	What the student does with model output	Example task
Remember	Verifies factual content against an authoritative source	Model states a solubility rule. Confirm or refute it against the data table, citing the source.
Understand	Explains the output in their own words	Model produces a balanced equation. Explain, in prose, why each coefficient is what it is.
Apply	Adapts the output to a changed condition	Model solves a stoichiometry problem at STP. Rework the solution for 40 degrees Celsius and 2 atmospheres.
Analyze	Locates and diagnoses an error	Prompt the model until it produces an incorrect mechanism. Identify the error and explain why it is wrong.
Evaluate	Judges output against explicit criteria	Given two model explanations of Le Chatelier's principle, judge which is better and defend the choice.
Create	Designs the interrogation itself	Construct a sequence of prompts that reliably produces a correct multi-step synthesis, and document what failed first.

The upper three levels are where this approach does most of its work. A student cannot outsource *finding the flaw* without understanding the chemistry well enough to know where flaws live.

Assignment template

Use this structure for any AI-integrated task. It fits on one page.

[Task title]

Learning objective. [What the student should be able to do afterwards.]

Bloom's level. [Analyze / Evaluate / Create]

AI use. Permitted and expected. You are not being tested on whether you used a model. You are being assessed on what you did with what it gave you.

What to submit.

1. **The prompt or prompts you used.** Copy them exactly, including ones that failed.
2. **The model's output.** Unedited, exactly as returned.
3. **Your analysis.** [The specific cognitive task — locate the error, adapt to the new condition, evaluate against the criteria.]
4. **Your corrected or extended work,** with reasoning shown.

Assessment. Items 3 and 4 carry the marks. Items 1 and 2 are required for the submission to be complete but are not themselves graded.

A note on honesty. If the model happened to be correct and you could not find an error, say so and explain how you verified it. That is a valid and creditable outcome.

That final note matters more than it looks. Without it, students invent errors to satisfy the assignment.

Worked example — chemistry, Analyze level

Diagnosing a flawed reaction mechanism

Objective. Identify errors in electron-pushing notation and explain why they violate mechanism rules.

Task. Ask a generative model to give the full mechanism for the acid-catalysed hydration of an alkene of your choosing. Ask it to describe every curly arrow.

Then:

1. Reproduce the mechanism as the model gave it
2. Identify at least one error — in arrow origin, arrow destination, intermediate stability, or a missing step
3. Explain why it is wrong, referring to the rule it breaks
4. Give the corrected mechanism with your own arrow descriptions

If the model's mechanism is correct throughout, demonstrate that by explaining why each step is justified, and note what you tested.

Marks. Error identification and explanation, 40 percent. Corrected mechanism, 40 percent. Quality of reasoning, 20 percent.

Worked example — physical science, Apply level

Adapting a stoichiometric calculation

Task. Have a model solve a gas stoichiometry problem at standard temperature and pressure. Then rework the same problem at 40 degrees Celsius and 2.0 atmospheres.

Submit. The original prompt and output, your reworked solution with all steps, and a short statement of which parts of the model's method survived the change in conditions and which did not.

Rubric

Criterion	Excellent	Adequate	Insufficient
Engagement with output	Identifies substantive errors or confirms correctness with genuine verification	Identifies surface errors only	Accepts output uncritically or asserts an error without support
Disciplinary reasoning	Explains using correct principles and terminology	Reasoning present but incomplete	Reasoning absent or incorrect
Corrected work	Complete and correct, with steps shown	Mostly correct, gaps in working	Incorrect or not attempted
Documentation	Prompts and outputs supplied in full, including failures	Supplied but partial	Missing

Implementation notes

Say it in the syllabus. Students need to know AI use is expected in these tasks and what the boundaries are elsewhere. Ambiguity produces anxiety and, ironically, more concealment.

Start with one assignment. Do not convert a whole course. Take one existing task, rebuild it at the Analyze level, and see what the submissions look like.

Expect the conversation to change. The most reliable observation from instructors using this approach is not a change in scores but a change in how students talk about the material. They begin arguing with the model rather than trusting it.

Do not require a specific tool. Students have unequal access to paid models. Any generative model should be acceptable, and a student who works the problem without one has not violated anything.

Accessibility. Model output pasted into a document carries no structure. Require students to submit output as text rather than screenshots, both so it is readable by assistive technology and so you can search it.

What this approach does not solve

It does not stop a determined student from having a model write the analysis as well. Nothing does.

What it does is remove the *incentive*, because the analysis is harder to fake convincingly than the answer was, and a fabricated analysis is usually obvious to an instructor who knows the material. That is a better position than detection software.

Feedback on this model is welcome and will be credited in future versions.